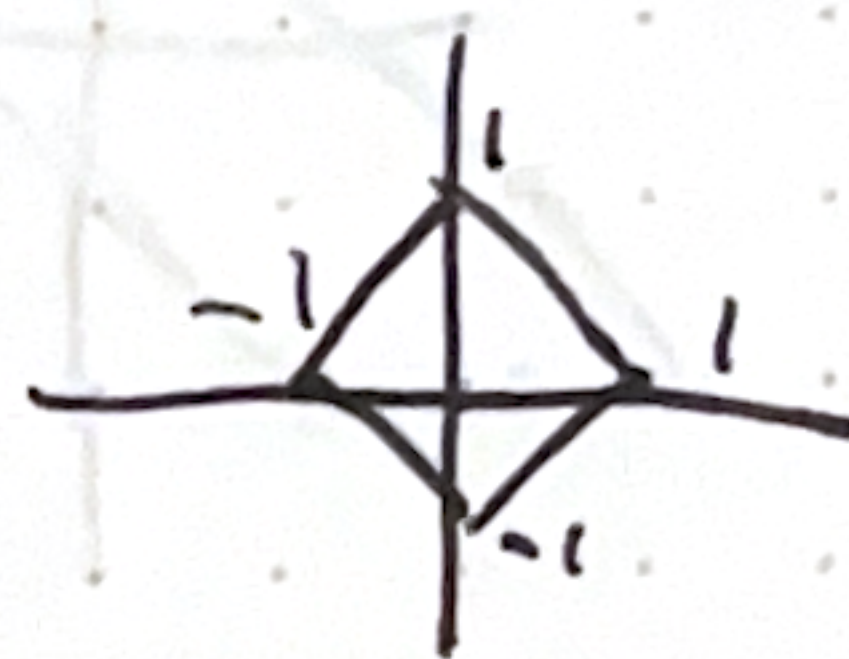


1.) $A = \begin{pmatrix} 1 & 2 \\ 0 & 2 \end{pmatrix}$

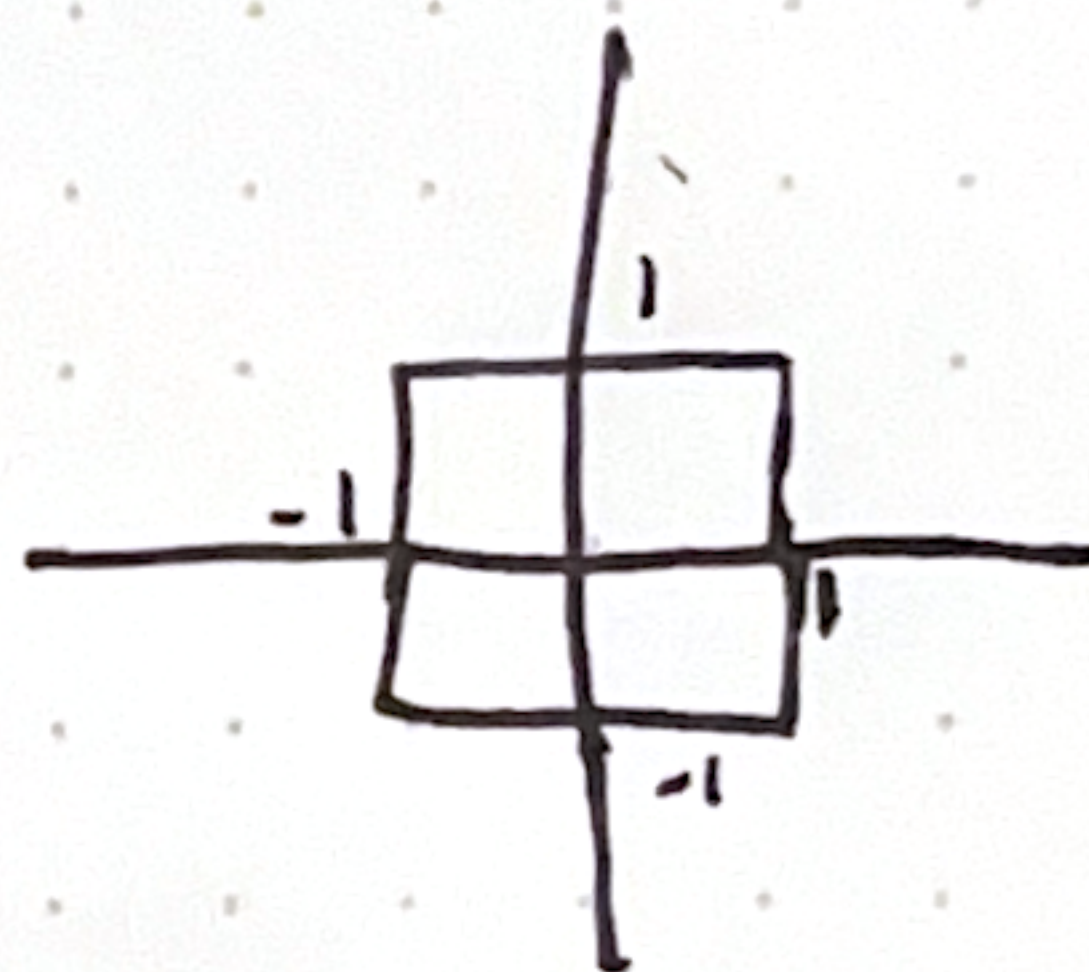
In $\mathbb{R}^2$, $\|x\|_1$ is a diamond



$\|x\|_2$ is a circle



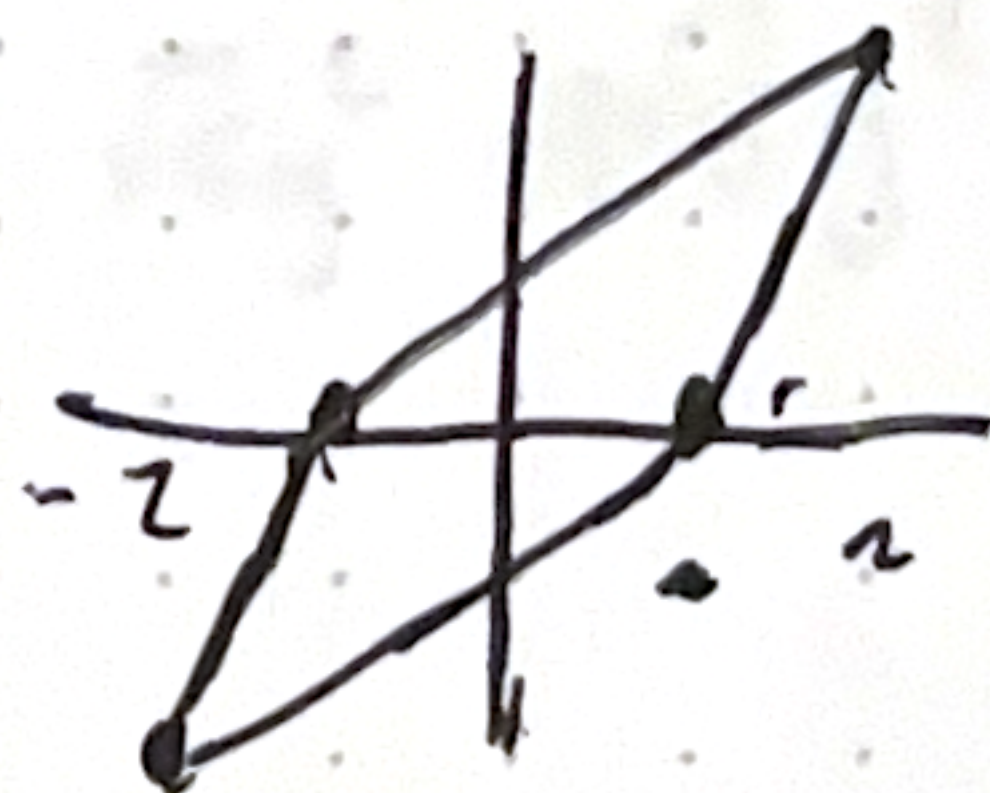
$\|x\|_\infty$ is a square



A is a simple 2D transformation matrix that stretches in the y direction by $k=2$ & shears parallel to the x direction by 2...

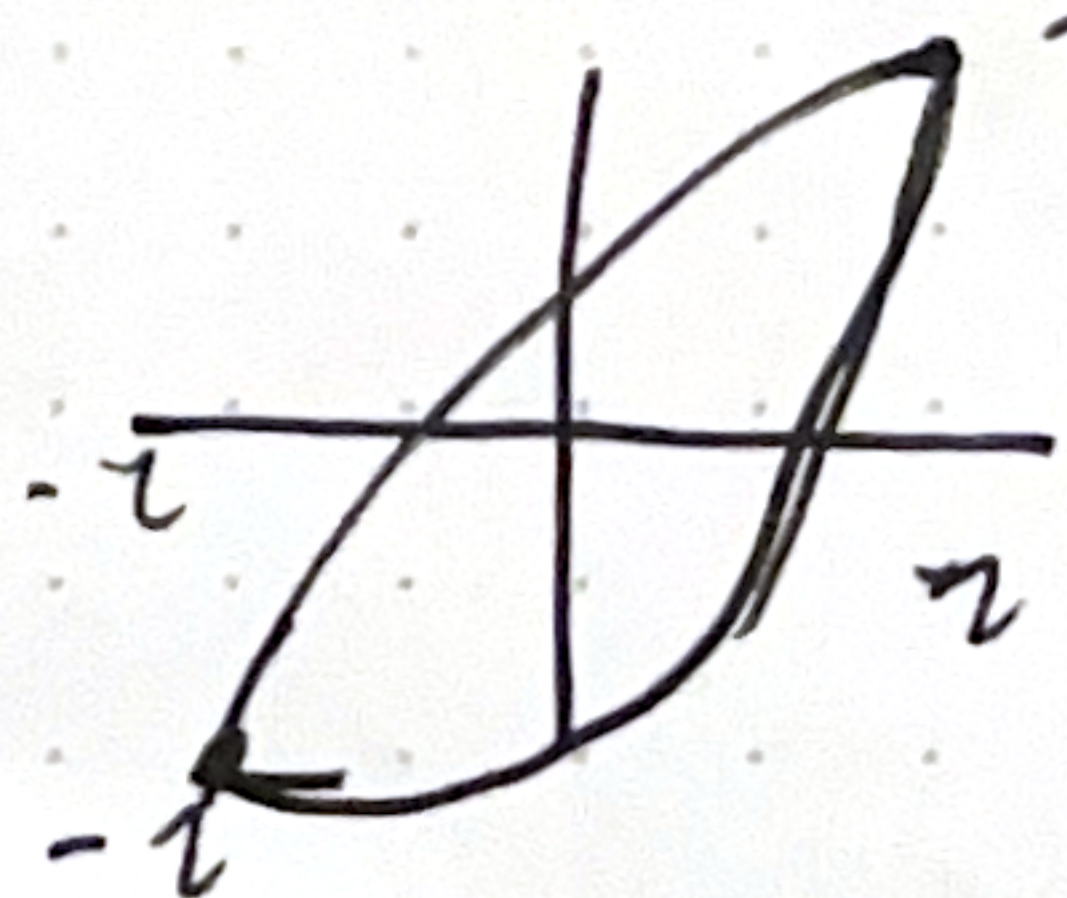
So, the ~~new unit ball~~ transformed unit balls become

$\|x\|_1 \rightarrow$



a stretched diamond

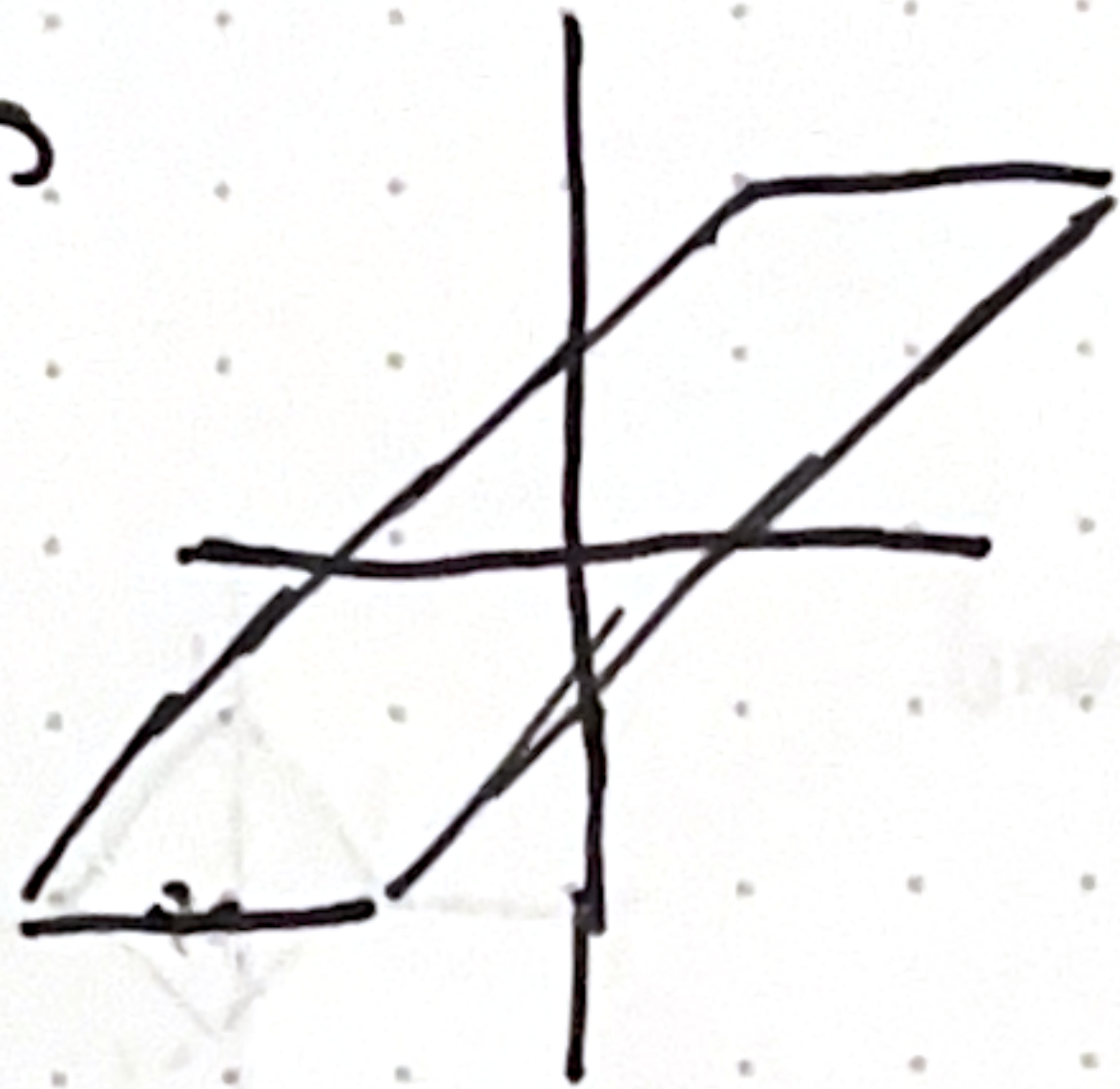
$\|x\|_2 \rightarrow$



an ellipsoid,

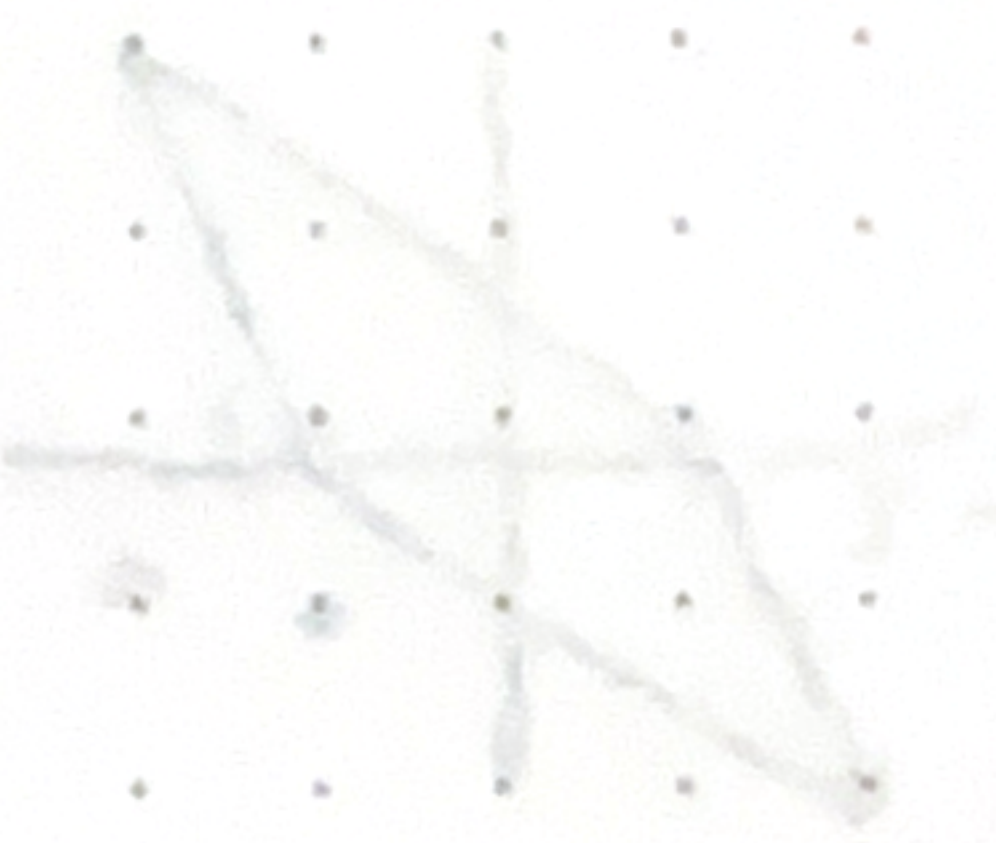
$11 \times 11_3 \rightarrow$

A parallelogram



A is a simple 2D coordinate system. The x-axis is horizontal and the y-axis is vertical. The origin is at the center of the coordinate system.

The x-axis is labeled 'x' and the y-axis is labeled 'y'. The origin is labeled 'O'.



4) Consider the surface $xy + 2xz = 5\sqrt{5}$; $(x, y, z) \in \mathbb{R}^3$

a) Find the coordinate integers affiliated w/ the min distance from a point on the surface to the origin

- want to minimize distance to origin, $d = \sqrt{x^2 + y^2 + z^2}$
 ↳ we can actually just minimize $x^2 + y^2 + z^2$ to simplify our lives.

use Lagrange multipliers ... $\nabla f = \lambda \nabla g$, where $g =$

$$f = x^2 + y^2 + z^2$$

$$g = xy + 2xz - 5\sqrt{5}$$

$$\rightarrow \begin{aligned} 2x &= \lambda (y + 2z) \\ 2y &= \lambda (x) \\ 2z &= \lambda (2x) \\ \& \quad xy + 2xz &= 5\sqrt{5} \end{aligned}$$

Solve the system ...

$$\lambda = \frac{2y}{x}, \quad \lambda = \frac{2z}{x} \rightarrow z = 2y$$

$$\rightarrow 2x = \frac{2y}{x} (y + 2z) \rightarrow 2x = \frac{2y}{x} (y + 4y) \rightarrow 2x^2 = 2y^2 + 8y^2$$

$$\rightarrow x^2 = y^2 + 4y^2 \rightarrow x^2 = 5y^2 \rightarrow x = \pm\sqrt{5}y$$

Constraint

$$xy + 2xz = 5\sqrt{5} \rightarrow \sqrt{5}y^2 + 4\sqrt{5}y^2 = 5\sqrt{5} \rightarrow 5\sqrt{5}y^2 = 5\sqrt{5}$$

$$y = \pm 1$$

$$\text{If } y = 1, \quad x = \pm\sqrt{5}, \quad z = 2$$

$$\text{If } y = -1, \quad x = \pm\sqrt{5}, \quad z = -2$$

so let's find min distance

$$\begin{aligned} (\sqrt{5}, 1, 2) &= \sqrt{5^2 + 1^2 + 2^2} = \sqrt{10} \\ (\sqrt{5}, -1, -2) &= \dots = \sqrt{10} \end{aligned}$$

Min distance for all points is $\sqrt{10}$